1. Appendix

Proposition (The inclusion-exclusion principle)**.** *For finite sets $A_1, A_2, \ldots, A_M$,*

$$\left|\bigcup_{j=1}^{M} A_j\right| = \sum_{\varnothing \neq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j\in I} A_j\right|.$$

Proof. The $M = 1$ case is trivial, and $M = 2$ case is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. We proceed inductively, so for $M \geq 3$, assume the result for $M - 1$ sets, and let us prove it for M sets. We split up the sum on the right of our desired equation into sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $\left|\bigcup_{j=1}^{M-1} A_j\right|$. The second sum is $\sum_{\{M\} \subseteq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j\in I} A_j\right|$, which we rewrite by separating the $I = \{M\}$ term and reindexing the rest using $J = I \setminus \{M\}$ to get

$$|A_M| - \sum_{\varnothing \neq J \subseteq \{1,2,\ldots,M-1\}} (-1)^{|J|-1} \left|\bigcap_{j\in J} \left(A_j \cap A_M\right)\right|.$$

Again using our inductive hypothesis, this final sum equals $\left|\bigcup_{j=1}^{M-1} \left(A_j \cap A_M\right)\right| = \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. The entire right side of our desired equation therefore equals $\left|\bigcup_{j=1}^{M-1} A_j\right| + |A_M| - \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. By the $M = 2$ case, this is $\left|\bigcup_{j=1}^{M} A_j\right|$, as desired. $\square$